Flexible self−protection as evidence of pain−like states in house crickets

Manzi, Lynch, Allman, Latty, & White (2026) − Proc. R. Soc. B 293: 20260609

INSTAR v1.0

FOUNDATIONS

Subjects

Taxonomic ID, life stage, & sex
Acheta domesticus (house cricket); adults of both sexes

Source & culture history
Commercially reared; Petstock (Australia); founding stock and generations in captivity not recorded

Sample size & attrition
n = 80 adults (40 M, 40 F); all analysed; fully within−subjects design; sample size not formally justified

Procedures

Capture, transport, & handling
Crickets gently immobilized on a sponge for each treatment application; immediately transferred to observation arena; identical handling across all three treatments

Anaesthesia, analgesia, & invasive procedures
Sponge immobilization only; no anaesthesia or analgesia (noxious heat was the experimental treatment)

Containment & biosecurity
Not reported

Ethics & Compliance

Ethics review, permits, & conservation status
No formal ethics approval required in Australia for invertebrate research; welfare reasoning explicitly provided in lieu (see Ethics statement)

Humane endpoints & non−target impacts
No early termination triggered; all individuals completed the protocol without harm and lived out their natural lifespans in captivity

Welfare & 3Rs statement
Welfare reasoning given: stable environment, continuous food and water, probe set to avoid lasting harm, post−study return to housing

WELFARE DOMAINS

Nutrition

Diet, feeding, & water
Wheatgerm ad libitum; peaches in juice as combined food and water source; no pre−experimental fasting

Environment

Housing & abiotic conditions
Shared holding containers 40 x 40 x 100 cm; 12:12 L:D; 23 plus/minus 1 C. Test arenas 30 x 22 x 10.5 cm at ~100 lux (low−stress) or ~1000 lux (high−stress)

Acclimation
Not reported

Field site & collection
Not applicable

Health

Health monitoring
Crickets monitored throughout the trial period (daily checks of responsiveness and condition); no formal scoring rubric

Injury & mortality
No injuries or unexpected deaths; all individuals lived out their natural lifespans post−experiment with no lasting effects

End of study
Returned to housing containers after the study and lived out their natural lifespans under continued laboratory care with free access to food and water

Behaviour

Behavioural opportunities, enrichment, & disturbance
Arena context deliberately varied in cover, substrate and illumination as part of the stress manipulation; 10−min intertrial interval in individual holding containers

Affective state

Indicators & precautionary measures
Site−directed grooming of focal antenna as pain−like behavioural indicator; 65 C noxious probe set point chosen to elicit response while avoiding lasting tissue damage (per Gibbons et al. 2024)

Empty fields = item not reported · "Not applicable" = not relevant to this study